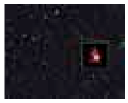
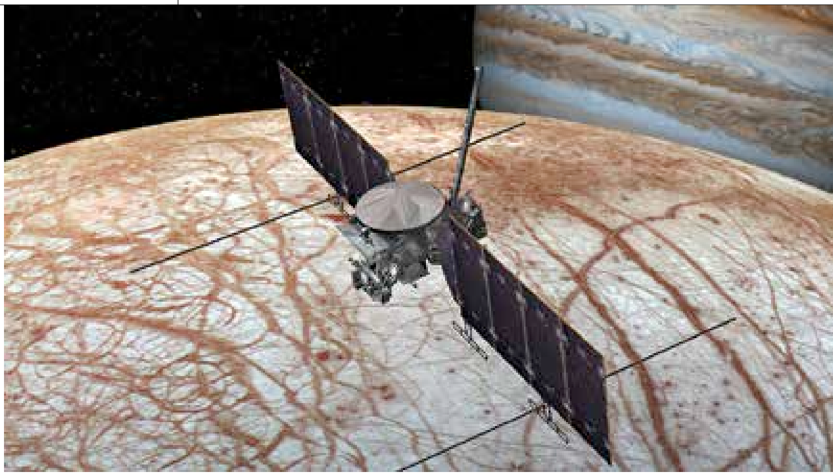
75 years of India's Independence



The Galaxy of the olden times
The James Webb Telescope discovers one of the oldest galaxies to ever be spotted by humans.
https://dgit.in/aug22-16



TO THE MOONS AND BACK!

In pursuit of habitable land beyond Earth

Priyanka Srinivas | feedback@digit.in

As part of NASA's Apollo programme, 12 astronauts visited the Moon between 1969 and 1972. Since then, humans have not returned, and the Moon remains as only planet that humans have ever visited. The Moon doesn't have an atmosphere similar to Earth's, and there isn't any wind or water to wash the Apollo astronauts' footprints away. The Moon also lacks tectonic activity, unlike Earth and Venus, hence its internal structure has been well preserved since its formation. Scientists now have the chance to comprehend how planets' interiors form. Our Moon is a historical encyclopedia containing information about ancient Earth. Similarly, exploring our solar system and understanding the planets around us involves closely examining the multiple moons that surround them. Jupiter, Milky Way's biggest planet, has many unexplored mysteries. Its moon or shall we say moons hold the answer.

JUPITER AND ITS SEVERAL MOONS

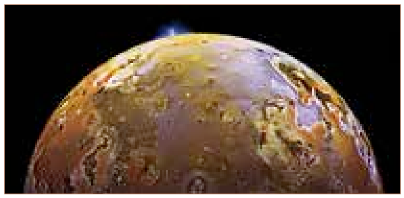
Jupiter has 53 named moons and more than 20 others awaiting formal identification. Currently, it is believed that Jupiter has 79 moons altogether. There are many fascinating moons around the globe, but the Galilean satellites—the first four moons identified outside of Earth—hold the greatest scientific curiosity.

THE GALILEAN MOONS

The Galilean satellites are named after Italian scientist Galileo Galilei, who made the first observation of Jupiter's four largest moons in 1610. Simon Marius, a German astronomer, claimed to have spotted the moons at about the same time, but because he didn't record his findings, Galileo is credited with making the discovery. Each of these big moons—Io, Europa, Ganymede, and Callisto—are unique worlds.

1. Io

The solar system's most volcanically active body is Io. Sulfur is present on Io's surface in a variety of vibrant forms. Io's slightly elliptical orbit around Jupiter causes "tides" in the surface to surge 300 feet (100 metres) high, creating enough heat to support volcanic activity and evaporate any remaining water. Hot silicate magma powers Io's volcanoes.



2. Ganymede

Ganymede is the largest moon in the solar system (bigger than the planet Mercury) and the only known moon to have its own internally generated magnetic field.

3. Callisto

Callisto's surface is severely cratered and old, providing a visible record of events from the solar system's early past. The extremely few tiny craters on Callisto, on the other hand, imply a low level of present surface activity.

4. Europa

The surface of Europa is primarily water ice, and there is evidence that it may be covering an ocean of water or slushy ice beneath it. Europa is thought to have double the amount of water as Earth. Astrobiologists are intrigued by this moon's potential for a "habitable zone." On Earth, life forms have been discovered thriving near subsurface volcanoes and in other harsh settings that could be parallels to what might exist on Europa.

THE EUROPA CLIPPER MISSION

About Europa

Europa is named after a woman who was kidnapped by Zeus in Greek